Workshop - Week 11

1. (a) Determine the length of $y = (x+4)^{3/2}$ between $0 \leq x \leq 4$.

(b) Determine the length of $x = \frac{y^4}{8} + \frac{1}{4y^2}$ between $1 < y < 2$.

(c) (Careful...tricky computations) Determine the length of $y = \ln(\sec x)$ between $0 \leq x \leq \frac{\pi}{4}$.

2. (a) Determine the surface area of the solid obtained by rotating $y = \sqrt{a^2 - y^2}$, $0 \leq y \leq \frac{a}{2}$ about the y-axis.

(b) Determine the surface area of the solid obtained by rotating $x^{\frac{2}{3}} + y^{\frac{2}{3}} = 1$, $0 \leq y \leq 1$ about the $y-$axis.

3. Determine whether the sequence converges or diverges. If it converges, find the limit.

(a) $a_n = \cos n$

(b) $a_n = \frac{1}{2n+3}$

(c) $a_n = (\frac{-1}{2})^n$

(d) $a_n = 2 + \frac{(-1)^n}{n}$

(e) $a_n = \frac{3^n}{n^3}$ (*Hint: Hospital rule to a well chosen function*)

(f) $a_n = (\frac{1 \times 2 \times \cdots \times n}{n^n})$ *(Hint: Compare with $\frac{1}{n}$.)*

4. Assume that the sequence converges. Find the limit of the sequence

$$\{\sqrt{1}, \sqrt{1+\sqrt{1}}, \sqrt{1+\sqrt{1+\sqrt{1}}}, \ldots\}$$